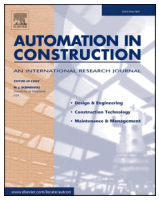
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Review

Deep learning-based data analytics for safety in construction



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ABSTRACT

Deep learning has been acknowledged as being robust in managing and controlling the performance of construction safety. However, there is an absence of state-of-the-art review that examines its developments and applications from the perspective of data utilization. Our review aims to fill this void and addresses the following research question: *what developments in deep learning for data mining have been made to manage safety in construction*? We systematically review the extant literature of deep-learning-based data analytics for construction safety management, including: (1) image/video-based; (2) text-based; (3) non-visual sensor-based; and (4) multi-modal-based. The review revealed three challenges of existing research in the construction industry: (1) lack of high-quality database; (2) inadequate ability of deep learning models; and (3) limited application scenarios. Based on our observations for the prevailing literature and practice, we identify that future research on safety management is needed and focused on the: (1) development of dynamic multi-modal knowledge graph; and (2) knowledge graph-based decision-making for safety. The application of deep learning is an emerging line of inquiry in construction, and this study not only identifies new research opportunities to support safety management, but also facilitates practicing deep learning for construction projects.

1. Introduction

Worldwide, safety in the construction industry is a challenge for practitioners [1,2,3]. For example, in the UK, approximately 30% of workplace accidents are from the construction sector [4]. According to the U.S. Bureau of Labor Statistics, the death toll of the construction industry in the United States climbed up to 1008 in 2018, i.e., 37 more than in that of 2017 [5]. In China, 904 people were killed during the construction of housing and municipal engineering projects in 2019 [6]. Despite an intensive attention being paid to safety management, the construction-related accident rate has remained to be high.

To manage safety in construction, emerging information technologies, such as Internet of Things (IoT) and computer vision, have been in place to monitor daily on-site activities [7,8,9,10]. As a result, substantial data (e.g., digital images and safety reports) have been recorded and stored, containing significant information for resolving such problems as unsafe behavior and structural defeats [7,11]. However, extracting useful information knowledge from such a large amount of data for safety management and decision-making is challenging [12]. Consequently, Deep learning, a subset of Artificial Intelligence (AI), has emerged. It refers to a method that is robust in automatically extracting sophisticated features at high levels of abstraction and, therefore, is promising for enabling a higher effective and efficient management of construction safety [13].

Over the past decade, deep-learning techniques have been further developed to enhance construction on-site safety [14,15,7,16]. For example, Fang et al. [15] and Ding et al. [14] developed deep learning and computer vision approaches to monitor and identify people's unsafe behavior. Despite the research above, there has been no state-of-the-art review with an attempt to address: *What developments in deep learning for data mining have been made to manage safety in construction*? [17,18,19]. Addressing the research question above requires a systematic review to garner an understanding of and examine deep learning and construction safety in terms of the data-mining perspective, so that headway can be made to engender and enact a robust focus on research and practice enhance of construction safety management [7,16].

This paper commences with an introduction of the data source for safety in construction (Section 2). Then, Section 3 introduces the development of deep learning in computer science. Next, the research approach used to form the study's line of inquiry is presented (Section

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4). In Section 5, we review the existing deep learning applications for safety in construction industry to provide a contextual backdrop of the research. Finally, the research challenges, recommendation of future research, and conclusion are discussed and presented.

2. Data source for safety in construction

Within the context of safety, we focus on unsafe behavior and relevant conditions that pervade on-site practice. It has been observed that 88% of the accidents that occurred on construction sites were as a result of unsafe behaviors, while 10% of that was attributable to unsafe work conditions (e.g., plant and structural monitoring). As addressed, AI requires different types of data (i.e., digital images and safety reports) useful for the management of safety. These include:

- *Image/video*: In the construction industry, video surveillance system is installed on the sites to record daily activities in real-time with permission. Additionally, the safety inspectors use cameras or phones to record hazards during their on-site inspection ((Fig. 1 (a)).
- *Safety report:* As noted from Fig. 1 (b), the safety report data (e.g., quality and safety reporting notice, weekly supervision form) are recorded by engineers who are experienced in documenting safety incidents, as is shown in Fig. 1(b).
- *Non-visual sensor data*: Having non-visual sensors (e.g., RFID, UWB, GPS) in place is also a common practice to monitor unsafe behavior and conditions. For example, positioning sensors are used to track people's location in construction sites. Likewise, sensors (e.g., displacement, deformation) are used to monitor surrounding settlement during excavation. This type of data is stored in numeric format, as shown in Fig. 1 (c).

3. Deep learning in computer science

The advent of deep learning algorithms has enabled a wide application of deep learning to various areas such as healthcare [20], self-driving cars [21], and language translations [22]. In line with the research question above, the deep learning techniques and the development of two prevailing applications (e.g., computer vision and natural language processing) are presented. We examine the deep-learning-related technologies in computer science, as they form the underlying basis for deep learning-based analytics in construction.

3.1. Deep learning techniques

Having benefitted from large-scale data training has advanced deep learning. However, in the industries where training data are lacking, the applications of deep learning are limited due to its rigorously required amount of inputted data. Transfer learning can leverage knowledge and skills learned in similar domains/tasks, and it is being applied as a common approach to overcome the absence of domain data sets [23,24]. Based on the representation of knowledge to be transferred, transfer learning algorithms can be categorized into: (1) instance-transfer; (2) feature-representation-transfer; (3) parameter-transfer; and (4) relational-knowledge-transfer [25].

Unlabeled data, essentially, is adequate or can be easily collected in many cases, while high-quality labeled data is normally limited and expensive to acquire. More importantly, the model's accuracy may not be significantly improved, though additional time is consumed with an enlarged training data set. In this regard, active learning has been developed, with an aim of iteratively selecting information samples from the unlabeled pool to minimize the cost of constructing datasets and then improve model performance [26,27]. Emerging from the application scenarios engenders three types of active learning: (1) membership query synthesis; (2) stream-based selective sampling; and (3) pool-based active learning [26]. Membership query synthesis actively generates new unlabeled instances to be queried. Stream-based selective sampling is based on a hypothesis that an unlabeled instance is observed at a time, and query selection is performed in order by identifying whether each instance is queried. Additionally, pool-based active learning performs query selection by prioritizing instances using acquisition functions and assumes that there are many unlabeled instances.

Data augmentation is another technique widely used in deep learning tasks. It deals with overfitting and contributes to improving the performance of model that is associated with unseen samples [28]. Data augmentation focuses on enhancing the value of limited data and can be divided into the supervised and unsupervised (data augmentation) methods. The supervised data augmentation adopts preset data transformation rules. It performs data augmentation depending on existing data, e.g., (1) single-sample data augmentation, which is based on the geometric or color transformation of the sample; and (2) multi-samples data augmentation, which integrates multiple samples to generate new samples. By contrast, the unsupervised data augmentation learns the distribution of data through the model, randomly generating samples that are consistent with the distribution of the training data set. Notably, a typical application of the unsupervised data augmentation algorithm is Generative Adversarial Network.

3.2. Deep learning for computer vision

Computer vision is an interdisciplinary field of science, which focuses on constructing computational models to capture contextual information from digital images or videos for automating the tasks of the human visual system [29]. Deep-learning-based computer vision has contributed to producing impressive results and even exceeded human-

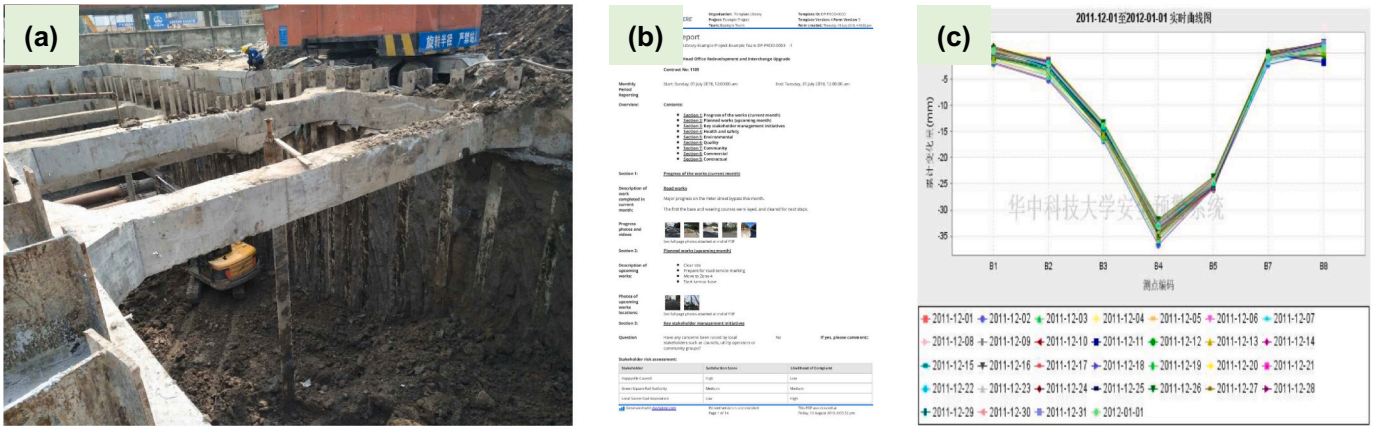


Fig. 1. Example of the data source in the construction industry.

level performance in some cases [30,31,32]. The typical tasks of computer vision include:

- *Object detection*: Object detection describes the ability to detect or identify objects in any given image correctly along with their spatial position in the given image. It is in the form of rectangular boxes (known as Bounding Boxes) within which the object it is bounded. Cases in point regarding object detection networks are relation to Faster R-CNN [33], F-RCN [34], SSD [35], YOLO [36] and RetianNet [37].
- *Object tracking*: Object tracking is to predict the size and location of the target in subsequent frames, given the size and location of the target in the initial frame of a video sequence. As a result, the position relationship of the object to be tracked in the continuous video sequence is established; thus, the complete trajectory of the object can be obtained. Typical attempts (tracking algorithms) that were developed based on deep learning include C-COT [38], DLT [39], FCNT [40], and MDNet [41].
- *Instance segmentation*: It aims to locate the individual objects and segment their boundaries (i.e., lines and curves) within a scene, regardless of whether they are under the same type. Well-known instance segmentation networks comprise Mask R-CNN [42], MaskLab [43], and YOLACT [44].
- *Pose/Activity recognition*: Pose/Activity recognition is used to understand human actions and behaviors by determining human joints' location from images, for example, CPM [45], Hourglass [46], CPN [47], and HRNet [48].

Table 1 presents the examples of deep-learning-based computer vision algorithms in the context of computer science.

3.3. Deep learning for natural language processing

Natural language processing (NLP) is a series of theoretically inspired computing techniques used to represent and analyze human language automatically [59,60]. The general pipeline of practicing NLP tasks is indicated in Fig. 2. Corpus is a collection of texts, which could be either standard public data sets or existing electronic texts. However, corpus preprocessing is time-consuming and laborious, including corpus cleaning, word segmentation, and stop words removal. Word embedding is adopted to convert words into vector representations prior to feature extraction.

The common tasks of NLP include:

- *Name entity recognition (NER)*: The NLP model is used to identify, extract and categorize entities from sentences. It comprises the extracting names of people, locations and things, which are derived from the text and placed under certain categories, such as person, location and time. For instance, BERT and BiLSTM-CRF are the applications of NER [61,62].
- *Text classification*: The NLP model is trained to classify documents/text reports according to specific attributes, e.g., subject, document type and time. The model extracts the features of the text and then matches them with the predetermined categories for classification. A popular case in point regarding text classification models is BERT [61].
- *Information retrieval:* The NLP model can locate the required information among unstructured data. An information retrieval system indexes a collection of documents and analyzes the user's query. Then, it compares the description of each document with the query and finally generates and presents the relevant results. The prevailing methods for information retrieval involve DRMM [63] and SNRM_PRF [64].
- *Text summarization*: An NLP algorithm can be used to create a shortened version of a text report with main points. Fundamentally, there are two general approaches for text summarization, i.e., abstractive and extractive summarizations. In the abstractive case, the NLP model engenders an entirely new summary in terms of phrases and sentences used in the analyzed text. Regarding the extractive summarization, the model extracts phrases and sentences from the existing text and groups them into a summary.

Table 1
Examples of deep learning-based computer vision algorithms in computer science.

Task	Algorithms	Performance	Reference
Object detection	SAPD	SAPD model can achieve a single-model single-scale AP of 47.4% on COCO database	Zhu et al. [49]
	FCOS	FCOS with ResNeXt-64x4d-101 achieves 44.7% in AP on COCO database	Tian et al. [50]
	FSAF	FSAF model can achieve a state-of-the-art 44.6% mAP on COCO database	Zhu et al. [51]
Object tracking	GM-Tracker	GMTracker achieves 63.8 IDF1 on MOT17 and 63.9 IDF1 on MOT16	He et al. [52]
	SiamMOT	SiamMOT (DLA-169) achieves 53.2 MOTA and 51.7 IDF1 on HiEve challenge	Shuai et al. [53]
	DyGLIP	DyGLIP achieves 84.6 MOTA and 39.9 IDF1 in S05 on AI City challenge database	Quach et al. [54]
Object segmentation	Dcnas	DC- NAS achieves 84.3% on Cityscapes, and 86.9% on PASCAL VOC 2012	Zhang et al. [55]
	DystaB	DystaB achieves 58.9 and 47.2 on DAVIS17 and Youtube-VOS database	Yang et al. [56]
Pose/Activity recognition	SimPoE	SimPoE achieves 56.7 MPJPE and 26.6 MPJPE on Human3.6 M and MPII- INF-3DHP dataset, respectively.	Yuan et al. [57]
	FCPose	FCPose achieves 64.8% AP^{kp} on the COCO dataset	Mao et al. [58]

4. Research approach

The systematic review, as addressed above, has been performed for this research. It is initially built on retrieving the relevant research papers from Scopus database. The use of this database is robust for optimizing search results for a systematic review [65]. A series of keywords, such as "deep learning", "construction safety" or "safety management" or "safety", and "construction", were inputted into the search engine of the selected database. Keywords were applied in the Title/Abstract/Keyword' (T/A/K) field of the Scopus. In this research, the filters were applied as follows: (1) peer-reviewed articles; (2) language: English; and (3) timespan: between 2012 and 2021.

To avoid missing influential papers, the Top 5 journals in terms of the number of their articles relating to deep learning-based construction safety were further searched to filter out the papers containing "deep learning" in their titles, or keywords or abstracts. These include *Automation in Construction (AIC)*, *Advanced Engineering Informatics (AEI)*, *ASCE Journal of Construction Engineering and Management (JCEM)*, *ASCE Journal of Computing in Civil Engineering (JCCE)*, and *Computer-Aided Civil and Infrastructure Engineering (CACIE)*. The filtered papers were then manually reviewed to determine their relevance. In summary, a total of 185 articles were identified for our next analysis.

Fig. 3 illustrates the distribution of the 185 bibliographic records from 2012 to 2021. There is a trend that the number of publications in this topic has significantly increased over the past five years. Accordingly, we conclude that deep-learning-based safety management has become and continues being a significant research scheme and, therefore, a plethora of scholarly works is expected in this area in the near future.

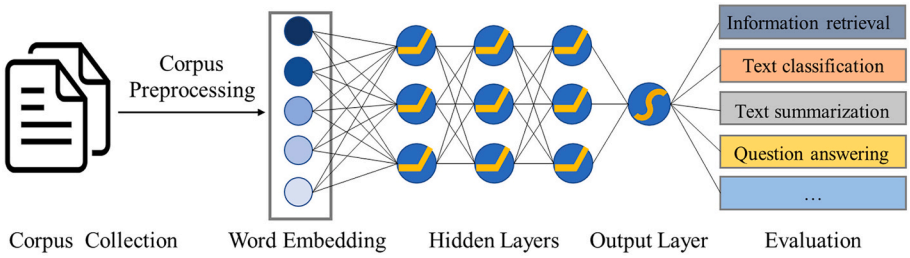


Fig. 2. General pipeline of NLP application tasks.

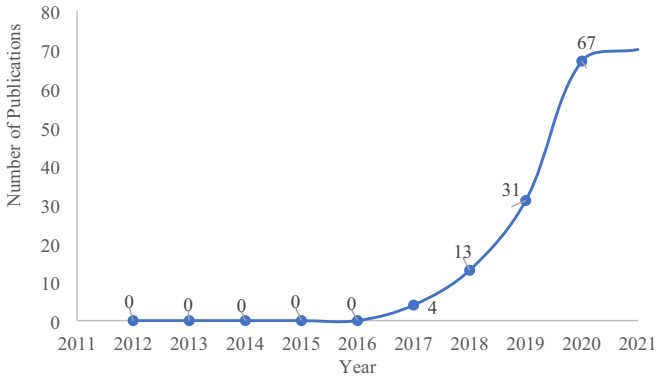


Fig. 3. The year profile of publications.

Research collaboration describes collaborative research works undertaken to achieve the common goal of producing new scientific knowledge. A collaboration analysis in this study was run to illustrate relationships from a micro to a macro perspective; that is, between scholars and institutions. Fig. 4 shows the co-authorship network of the bibliographic records of the observed period (2012–2021). As noted, there have been strong collaborative relationships between multiple institutions, e.g., the research institute of Prof. Heng Li, Prof. Lieyun Ding and Prof. Hanbin Luo.

Fig. 5 presents the results of the co-occurrence keywords. The minimum number of occurrences of a keyword is set to 5. The size of the nodes represents the frequency of the keywords, which can be split into two categories: (1) method-related (e.g., deep learning, convolutional neural network and computer vision); and (2) topic-related (e.g., construction safety, construction workers and structural health monitoring).

5. Deep learning-based data analytics for safety

As noted from the above sections, deep-learning-related techniques relate to data analytics. Therefore, they have the potential to accurately detect unsafe behavior and unsafe condition. Considering this point of view, we examine the progress of deep-learning-based data analytics for safety in construction.

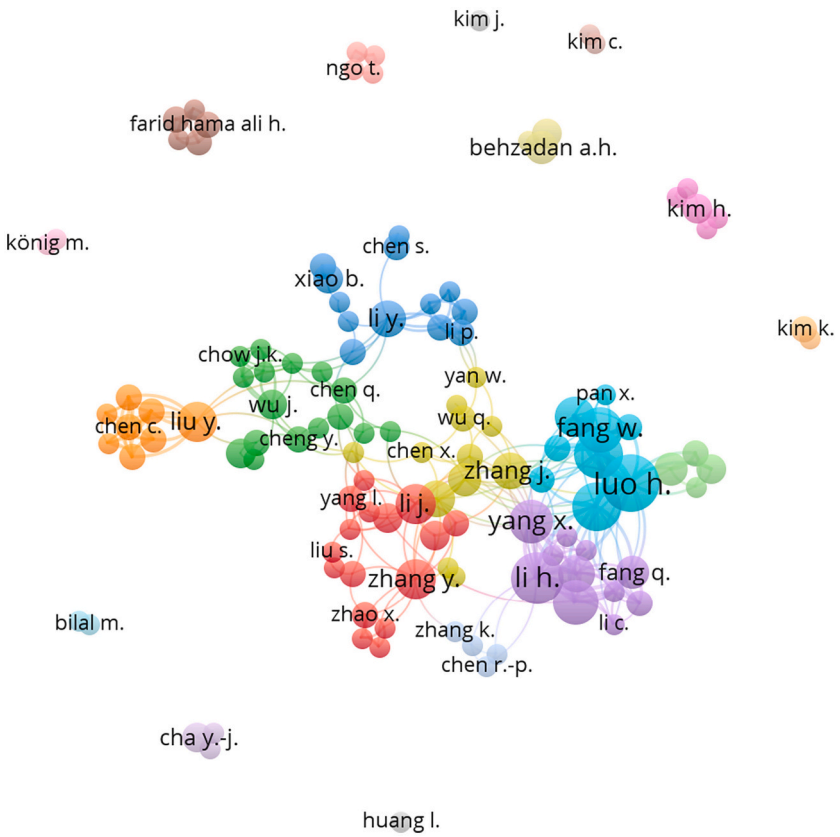


Fig. 4. Co-authorship network.

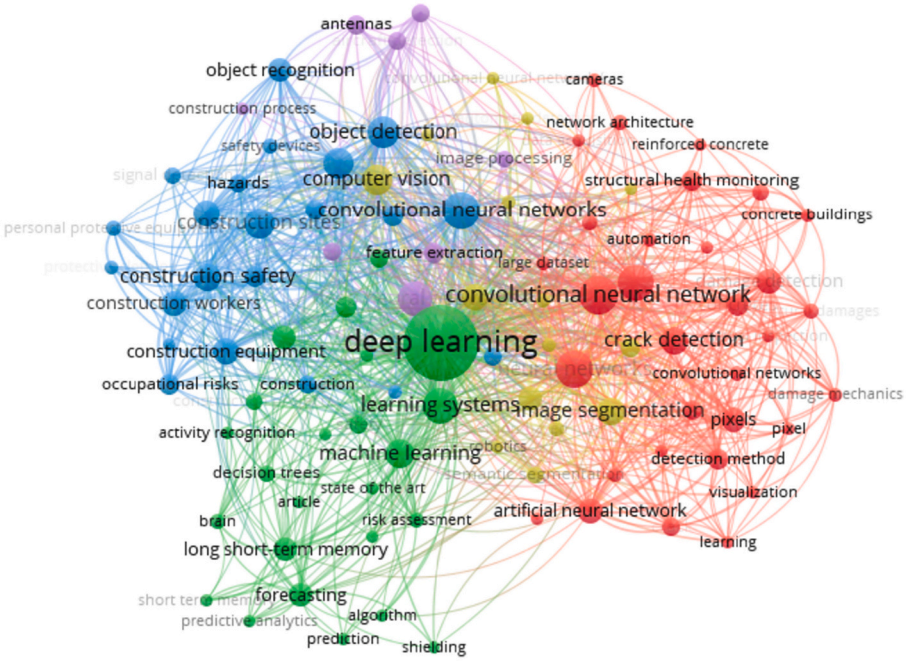


Fig. 5. Keywords co-occurrence network.

5.1. Image/video-based data analytics

Exploring the relevant literature has identified that previous studies were focused on using image/video to manage construction safety. They primarily concentrated on: (1) prevention of people's unsafe behavior; (2) structural health monitoring.

5.1.1. Prevention of people's unsafe behavior

Research has shown that approximately 88% of accidents are caused by people's unsafe behavior [66,67]. An accurate detection for people's unsafe behavior in real-time is thus pivotal for managing safety behavior. As demonstrated above, deep-learning-based computer vision approaches have been adopted by past studies to identify people's unsafe behavior automatically. According to the methods used for behavior monitoring, we categorized such studies into (1) object detection; (2) proximity measurement; (3) activity-based; and (4) semantic reasoning.

5.1.1.1. Object detection. Since studies suggest that a considerable number of injuries occurred as the result of the absence of personal protective equipment (PPE) [68], and detecting whether workers wear PPE as required has been the hotspot, including the detection of the hard hat [69,70,71], vest [69] and safety harness [72]. Emerging from Nath et al.'s research [69], for example, is related to three deep learning approaches that are built on You-Only-Look-Once (YOLO) V3 for determining if a person wears a hard hat, vest, or both properly. The result showed that the one-step end-to-end approach ensures the best performance with a mean average precision of 72.3% and a processing speed of 11 frames per second.

5.1.1.2. Proximity measurement. The improper spatial relationship on construction sites engenders risks for people who enter the hazard area. Fang et al. [73] utilized Mask R-CNN to detect people traversing structural support according to whether their masks overlap. Continuing with this theme, more studies have developed deep-learning-based computer vision methods to detect people entering dynamic hazard areas, such as the working area of heavy vehicles, which can lead to struck-by accidents [74,75,8,76]. For instance, Luo et al. [8] applied YOLO v2 to detect plant and people and then measured the distance between them based on perspective transformation. Similarly, Yan et al. [76] estimated the 3D spatial relationship between people and heavy vehicles using the 3D bounding box reconstructed from the video that is recorded by the monocular camera, aiming to address the spatial relationship distortion caused by two-dimensional image pixels-based estimation.

5.1.1.3. Activity-based. People-activity monitoring enables the application of automatic detection for unsafe behavior, which relies on mining spatiotemporal information in video frames [14,77,78,79]. For example, Ding et al. [14] applied a CNN model to extract the visual features from videos and utilized an LSTM model to sequence the learning features for unsafe behavior recognition, which outperformed the descriptor-based methods. In response to the characteristics of far-field surveillance videos, Luo et al. [78] adapted a deep action recognition method with a fusion strategy to recognize workers' actions and then proposed a Bayesian nonparametric hidden semi-Markov model to infer people's activities based on action sequences.

5.1.1.4. Semantic reasoning. The task-oriented model developed based on predetermined rules cannot identify multiple types of unsafe behaviors. This is because it is unable to accommodate the adjustments of safety regulations either. To solve this issue, the semantic-reasoning-based approach has been developed [80,81,82,83]. For example, Tang et al. [84] proposed a human-object interaction (HOI) recognition model to detect spatial relationships among objects in digital images so that hazards implied in the image can be determined and detected. Continuing with this, Fang et al. [80] combined computer vision algorithms with ontology to develop a knowledge graph to understand construction scene and then infer whether there is an existence of unsafe behavior. Likewise, Pan et al. [81] integrated zero-shot learning techniques with a knowledge graph to improve the accuracy and timeless knowledge graph updates. Table 2 summarizes prior key studies on computer vision and deep-learning-based unsafe behavior recognition.

5.1.2. Structural health monitoring

Structural health monitoring (SHM) is a strategy and process for damage identification and characterization of engineering structures [89]. Structural damage refers to the change of structural material parameters and geometric characteristics, which can lead to structural failure as well as significant safety accidents under unforeseen

Table 2
Examples of computer vision and deep-learning-based unsafe behavior recognition.

Task	Description	Author (Year)
Object detection-based	Three deep learning models were proposed to detect if workers are wearing hard hat and vest	Nath et al. [69]
	A Faster-R-CNN and deep CNN model was developed to identify workers and their harness	Fang et al. [85]
	A deep learning model for non-hardhat-use (NHU) detection is developed	Fang et al. [15]
	A real-time smart video surveillance monitoring system was developed to recognize people entering dangerous areas on a construction site	Luo et al. [8]
Proximity measurement-based	A computer vision method for UAV-assisted remote proximity monitoring was presented	Kim et al. [75]
	A real-time warning system using visual data is proposed to protect workers from potentially dangerous situations involving equipment operations	Son et al. [86]
Activity-based	A hybrid deep learning model is proposed to automatically recognize workers' abnormal actions	Ding et al. [14]
	Using deep learning approach to identify irregular construction activities from image data	Lin et al. [87]
Semantic reasoning-based	A knowledge graph integrating computer vision with ontology is developed for identifying hazards on construction sites	Fang et al. [80]
	The framework integrates computer vision and ontology is developed for semantically reasoning hazards and mitigations	Wu et al. [88]

circumstances. As shown in Table 3, existing literature is replete with studies applying deep learning to SHM.

Extensive efforts have been put into vision-based damage detection, for example, crack detection [93,94,91,95,96,104,97,98,99], pothole detection [100,101,92], corrosion detection [102], bolt-loosening detection [103] and sewer pipe defect detection [104,106,105]. Such works above have enabled developing a taxonomy comprising: (1) classification to determine whether there are defects in the image; (2) object detection to locate the defects in the image; and (3) segmentation to measure the geometric size of defects.

To take crack detection as an example, attempts have been made to explored deep learning models to determine whether an image/image patch contains cracks. For instance, Gopalakrishnan et al. [23] used the VGG-16 DCNN, pre-trained on a massive ImageNet database with transfer learning to search out the images containing cracks. Kang and Cha [91] introduced a deep CNN with a sliding window technique to analyze video data collected from UAVs. In particular, an autonomous UAV method using ultrasonic beacons and a geotagging method for the localization of damage were proposed.

Other researchers applied semantic segmentation techniques to provide a pixel-wise prediction of cracks in the images. Emerging from Bang et al. [93] is a deep convolutional encoder-decoder network, which is useful for identifying road cracks in black-box images at the pixel level. The encoder is composed of the convolutional layer of the residual network to extract crack features, and the decoder consists of the deconvolution layer to locate cracks' edges in the input image. In addition, the estimation of structural strength based on computer vision techniques has also received attention. Wang et al. [108] demonstrated the feasibility of the faster region-based convolutional neural network to evaluate the residual strength of corroded textile-reinforced concrete, i.e., a mean average precision of 98.98%.

While there has been a significant achievement of computer-vision-based people's unsafe behavior detection and SHM, issues associated with identifying multiple defects or multiple unsafe behaviors in real projects remain unaddressed because of "noise" in the construction industry.

Table 3
Prior works on deeplearning-based structural health monitoring.

Application/ Task	Description	Type of data	Reference
Crack detection	1) Whether there are cracks	Image, video	[23,90]
	2) Locate cracks in the image	Image, video	[91,92]
	3) Geometric measurement	Image, video	[93,94,95,96,97,98,99]
Pothole detection	1) Locate potholes in the image	Image, video	[92]
	2) Geometric measurement	Image, video, point cloud	[100,101]
Corrosion detection	1) Locate corrosion in the image	Image, video	[102]
Bolt-loosening detection	1) Locate bolts in the image and estimate the rotational angles	Image, video	[103]
Sewer pipe defect detection	1) Whether there are defects	Image, video	[104]
	2) Locate defects in the image	Image, video	[105]
	3) Geometric measurement	Image, video	[106]
Structure strength estimation	1) Concrete strength evaluation	Image	[107,108]
Structure damage diagnosis	1) Provide fined-detailed defect information for structure	Vibration signal	[109,110,111,112]

5.2. Text-based data analytics

Safety reports record valuable sources of information about how and why an event occurred. The existing attempts to analyze such reports depend on manual processing, while the unstructured free-text format could make the work be lengthy and challenging. With the development of deep learning in NLP, there is a trend to explore deep learning techniques for automatic safety report analysis. The examples of typical application scenarios are shown in Table 4.

Table 4
Prior works on deep learning-based safety report analysis.

Application/ Task	Description	Method	Reference
Accident causes analysis	1) Classify accident causes	Word2Vec skip-gram model and a hybrid structured deep neural network	[113]
		Convolutional bidirectional long short-term memory model	[114]
	2) Identify the most impactful hazards and behaviors	CNN and hierarchical attention networks	[115]
Hazard classification	1) Classify hazard types	BERT	[116]
		Word embedding, CNN, and LDA	[117]
		LDA, CNN, Word Co-occurrence network and word cloud technology	[118]
Accident information extraction	1) Critical information extraction from accident reports	BiLSTM-CRF	[119]

The analysis of accident causes extracted from accident reports is useful for understanding, predicting, and preventing the occurrence of construction accidents. In this instance, Zhang et al. [114] proposed a convolutional bidirectional long short-term memory-based method to classify the construction accident narratives according to the type of the accident cause. Moreover, Baker et al. [115] applied Hierarchical Attention Networks to learn valid injury precursors from accident reports to identify the most significant hazards and behaviors in a fully automated and data-driven manner. In contrast, classifying safety reports based on hazard types has received attention [116,117,118]. For example, Zhong et al. [118] applied a Latent Dirichlet Allocation algorithm to identify hazard topics, then used a Convolution Neural Network algorithm to classify hazard records, and finally generated a Word Co-occurrence Network to determine the interrelations between hazards and exploited Word Cloud technology to provide a visual overview of hazard records. In addition to the classification task focusing on the singular dimension, Feng and Chen [119] adopted the BiLSTM-CRF model to extract critical information from accident reports, including date, location, accident causes, type of accident fatality, and injury situations.

Despite significant contribution above made to the text-based information extraction for safety, with the following issues, such as open information extraction, long text-based information extraction and dynamic information update, are still being questioned. Further, using information that has been extracted for safety decision-making is being challenging for research.

5.3. Non-visual sensor-based data analytics

Managing construction safety through the use of non-visual sensor is one of foci in the existing literature. The research scopes of relevant studies include: (1) prevention of plant accident; and (2) structural health monitoring.

5.3.1. Prevention of plant accident

Although the plant-related accidents can result in severe casualties and significant economic loss, the existing research on applying deep learning to manage such issues is limited. In essence, several studies have made headways to the operation status control of the shield machine [120,121,122,123]. For example, Zhou et al. [123] proposed a hybrid deep learning model composed of the wavelet transform, CNN, and LSTM, to predict shield machine's attitude and position during Tunnel Boring Machine (TBM) excavation. Their model overwhelmed the three widely used predictive models (ARIMA, LSTM, and WLSTM) in terms of prediction accuracy. Subsequently, Zhang et al. [122] identified a cyber-physical system (CPS)-based hierarchical autonomous control scheme to support pressure balance control of slurry shield tunnel boring machines. Emerging from Zhang et al. [122]'s work is a deep learning model for the coordination-level controller. However, their result showed that its performance was not better than an autonomous control system based on the proposed switched model predictive controller.

In the case of plants, there has been a paucity of research examining their location and status. This is due to the difficulties in the identification of their mechanical parameters and the extraction of attributes. Moreover, AI-based plant status prediction using historical and real-time data remains a grand challenge in real projects.

5.3.2. Structural health monitoring

Vibration signal is another data source widely used for SHM, and the underlying assumption is that the dynamic characteristics and responses will be changed when there damages exist. In this instance, Sajedi and Liang [112] applied a fully convolutional encoder-decoder neural network for vibration-based SHM technique and performed semantic damage segmentation in a grid environment framework available for large-scale SHM. The proposed model achieved an accuracy of 96.3% and 93.2% in locating damage and classifying sixteen damage mechanisms, respectively.

5.4. Multi-modal-based data analytics

Multiple data fusion is the process of integrating data from different sources to produce more consistent, accurate, and useful information than that collected from a singular 'supplier'. Some research has tended to focus on how multi-modal can be fused and used to manage safety in construction, attempting to leverage reports to improve the accuracy of managing safety. For example, Fang et al. (2021) applied a matching approach by fusing image features and text features to detect people's unsafe behavior from images [124]. While headway is being made to manage safety by leveraging reports to improve the accuracy, multi-source data has not been fully utilized and fused for safety.

6. Research challenges

Deep learning has been a important technique to support safety management in the construction industry. Nonetheless, there is an existence of challenges that hinder its applications.

6.1. Lack of high-quality database

Deep learning for safety management in the construction industry requires an extensive database for model training. However, the current database is limited to specific tasks such as worker and excavator detection [125,126]. One of the main challenges hindering the creation of a high-quality database is about establishing the ground-truth label.

The ground-truth labels are critical to the performance of deep learning methods. Nevertheless, it is difficult to obtain such high-quality labels due to the limited availability of experienced safety engineers as well as huge manual ground-truth labeling workloads. For example, the ground-truth label of psychological signals used for occupational safety requests people to complete questionnaires, which can be subjective and error-prone [127,128]. Likewise, the supervised deep learning-based computer vision algorithms for hazard identification need thousands of images and corresponding labels. Such training processes requires human experts to label images and indicate the relationship between construction resources (e.g., worker, equipment and environment in the scenes), which may result in 'false alarms' in those manual labels. The false labels can significantly mislead the machine learning process and generate an issue that computer vision models produce false hazard detections.

Collecting quality data in complex and dynamic working conditions is another challenge for high-quality database creation. The harsh work environment and people's perspiration and movement can adversely affect the signal quality of the physiological sensors for occupational health and safety monitoring. Put simply, a model that performs well in the laboratory may not effectively work on a construction site in practice.

Strategies, such as data augmentation (e.g., noise addition, generative models) and transfer learning, can contribute addressing and resolving the above issues. Further research into augmentation techniques would produce more extensive data sets, thereby enhancing the performance of deep learning. In addition, semi-supervised learning or unsupervised learning is also a promising way to accommodate the challenges above, because they require smaller samples for training. As such, future work is needed to develop new semi-supervised learning or unsupervised learning algorithms for safety.

6.2. Limited ability of deep learning models

Extant studies tend to use a relatively small database to train deep learning. Therefore, the generalization of the models needs to be enhanced. Generalization describes the ability of a trained machine

learning model to predict unseen samples accurately during the model testing phase [129]. Thus, a good generalization is needed for any practical deep learning algorithms. Normally, the training and testing samples are different, and the performance of a deep learning model during testing stage is worse than that in the training process.

Multiple factors may determine the generalization performance of deep learning models, such as cross-site and cross-device. For example, there are many wristbands for psychological signal collection on the market. Although many of them can measure PPG signals, the signal quality varies by brand. Even for devices from the same company, the quality of signals is varied with the generation of different models. Thus, signal processing and machine learning approaches need to handle cross-device variations effectively. Transfer learning techniques can be used to alleviate the above limitations by making use of data or machine learning models from an auxiliary domain or task.

Compared with traditional machine learning algorithms, the superiority of deep learning lies in the elimination of feature engineering design. However, deep learning models have been identified as being "black boxes". In this instance, it is difficult for engineers to understand the model's learning processes and how detection is made and to identify the parameters that need to be adjusted for improving detection accuracy. To address the above challenges, there is a need for deep learning models to integrate engineering knowledge for characterizing engineering.

6.3. Limited to specific applications

Whilst significant development has been made to identify hazards using deep learning, it is essential to realize that these approaches are being specific for some certain tasks of safety only, such as defect (e.g., crack) detection, or not wearing PPE detection. Put simply, a wide range of problems has not been well addressed by using deep learning techniques. With the advancement of big data, cloud computing, deep learning, and other informatics technologies, developing new methods is needed to discover knowledge from multi-source data and train them to detect on-site safety issues in construction within a wider context.

7. Recommendation for future work

Addressing the challenges discussed above requires future research. Noteworthy, existing studies on deep-learning-based safety are detection-focused and have been limited to certain safety-related problems. With this in mind, future research is needed to be conducted within a wider context so that outcomes generated from deep learning can be learned by site managers to better manage safety issues and then identify and take actions to control and minimize them. Accordingly, we have proposed a framework to illustrate how deep learning can be further developed and utilized to mine data for safety management in the construction industry (Fig. 6).

7.1. Construction of dynamic multi-modal knowledge graph

Currently, safety-related information is stored by different stakeholders in the form of heterogeneous data (e.g., images, videos and texts). The data is scattered in multiple systems with diverse, complex and isolated characteristics. The value of a single data is not prominent. To address the emerging challenges, future research should focus on developing new solutions to integrate and use multi-source data for knowledge extraction and decision-making.

One of the feasible solutions for data integration is developing a multi-modal knowledge graph, where entities and their interdependency (relationships) are represented and captured as nodes and edges, respectively [130]. Both the nodes and edges in a knowledge graph are high-dimensional. As addressed above [130], Fang et al. [80] are claimed as a pioneer to embed computer vision into ontology to establish a knowledge graph for unsafe behavior recognition that is underpinned by rule-based reasoning. Multiple unsafe behaviors in the scene can be identified at one time, however, the developed knowledge graph is small-scale with very limited reasoning ability.

As noted from Fig. 6, the first step for constructing a multi-modal knowledge graph is to develop a domain ontology. We have identified that safety codes, historical safety reports, and expert experience are pivotal for enabling the ontology (Fig. 6).

The second step is of extracting knowledge from multi-source data. Deep-learning-based computer vision model is robust in extracting knowledge (from images), understanding the objects, object-object interaction, and implying unsafe events (from images/ videos). Similarly, the NLP technique should be developed to identify entities and entity-entity/attribute relationships. Then, different types of knowledge graph can be identified from the extracted knowledge accordingly. The well-defined entities enable filtering relevant information from multi-source data and constructing precise knowledge graphs [130].

The final stage of the Step 2 relates to the fusion of different knowledge graph. Put simply, it is regarding how to link different knowledge graphs and fuse them. We have constructed different knowledge graphs based on the previous discussion, such as the image/ video knowledge and safety report-based event knowledge graphs. Additionally, the update of the multi-modal knowledge graph should also be considered, as multi-source data can be collected using sensors (e.g., CCTV) and manual inspection daily. Hence, accurately updating the knowledge graph and reducing the "noise" of the data will be research challenges in our future work.

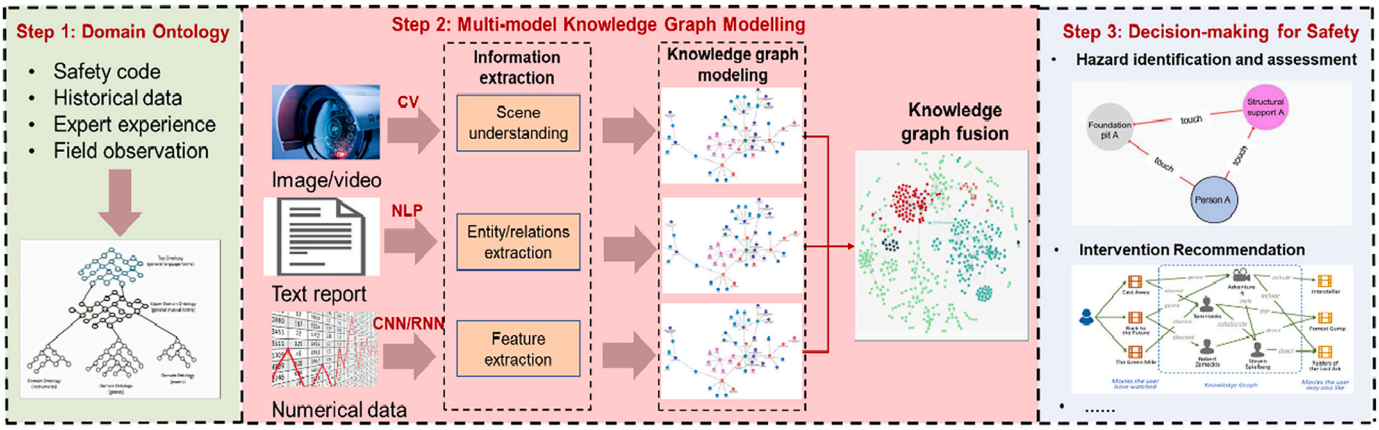


Fig. 6. Proposed framework.

7.2. Knowledge graph-based decision-making for safety

With an accurate multi-modal knowledge graph, the final step of the proposed framework above (Step 3) is with regard to making effective and efficient decisions to reduce or prevent accidents. Decision-making in the construction sites needs to consider different relief stakeholders (e.g., construction unit, supervision). In addition to identifying and assessing hazards from as-built multi-modal knowledge graph in construction site, future work on decision making needs to focus on: (1) dynamic prediction of hazards on construction sites; (2) recommendation of knowledge for hazards prevention; (3) recommendation of knowledge for hazard intervention (e.g., worker safety training).

We suggest that hazards on construction sites can be predicted in an early manner. Noteworthy, the characteristics and patterns of an unsafe event can be noted before it occurs. For example, in the case where a person enters the excavator dangerous area, their direction will be toward the excavator, which is subject to a closer distance to the worker. Based on the as-built multi-modal knowledge graph, the characteristics and patterns of a hazard can be extracted and analyzed so that hazards can be predicted in an early manner. Nevertheless, the following research challenges need to be considered: (1) how can an event's characteristics and patterns be defined before it occurs? (2) how to predict the hazard according to the extracted information? and (3) how early can the hazard be predicted?

Knowledge for different stakeholders can also be recommended to take actions to prevent accidents as per the as-built multi-modal knowledge graph. Through the consideration for the type of hazards, the multi-modal knowledge graph can be used to recommend possible solutions to prevent hazards. In this instance, it is essential for future research to address three questions: (1) how to define the hazard from the perspective of knowledge? (2) how to design a recommendation system and then recommend knowledge to the stakeholders based on the multi-modal knowledge graph? and (3) how to evaluate the effectiveness and feasibility of the recommendation system on hazards solution?

8. Conclusion

We have placed a systematic review in this paper to understand the developments of the deep-learning-related studies and their applications to construction safety, focusing on four key areas according to different types of data: (1) image/video-based; (2) text-based; (3) non-visual sensor-based; (4) multi-modal-based. Furthermore, our review has provided an insight into the challenges confronted by research and practice. These involve: (1) lack high-quality database; (2) inadequate ability of deep learning models; and (3) limited to specific applications.

To further manage safety through the use of deep learning, we have identified that our future study needs to focus on utilizing multi-source data and extracting knowledge from the data. In doing so, there is also a need of paying attention to: (1) the construction of dynamic multi-modal knowledge graph; and (2) knowledge graph-based decision-making for safety management. To this end, the contributions of our research are threefold, as we have: (1) provided novel insights into examining the development of deep learning for studying safety in construction from the perspective of data utilization; (2) identified the research challenges of adopting deep learning; and (3) presented new research opportunities to support and facilitate deep learning-based data analytics for safety in construction.

Declaration of Competing Interest

This manuscript has not been published or presented elsewhere in part or in entirety and is not under consideration by another journal. We have read and understood your journal's policies, and we believe that neither the manuscript nor the study violates any of these. There are no conflicts of interest to declare.

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References

[1] S. Guo, L. Ding, Y. Zhang, M.J. Skibniewski, K. Liang, Hybrid recommendation approach for behavior modification in the Chinese construction industry, J. Constr. Eng. Manag. 145 (6) (2019) 04019035, https://doi.org/10.1061/(ASCE)CO.1943-7862.0001665.

[2] P.E.D. Love, L. Ika, B. Luo, Y. Zhou, B. Zhong, W. Fang, Rework, failures, and unsafe behavior: Moving toward an error management mindset in construction, IEEE Trans. Eng. Manag. (2020) 1–13, https://doi.org/10.1109/TEM.2020.2982463. In press.

[3] P.E. Love, P. Teo, J. Smith, F. Ackermann, Y. Zhou, The nature and severity of workplace injuries in construction: engendering operational benchmarking, Ergonomics 62 (10) (2019) 1273–1288, https://doi.org/10.1080/00140139.2019.1644379.

[4] Health and Safety Executive, Workplace Fatal Injuries in Great Britain 2021, 2021. http://www.hse.gov.uk/statistics/pdf/fatalinjuries.pdf (accessed Feburary 14, 2022).

[5] U.S. Bureau of Labor Statistics, Census of Fatal Occupational Injuries. https://www.bls.gov/iif/oshwc/cfoi/cftb0330.htm, 2019 (accessed Feburary 14, 2022).

[6] Ministry of Housing and Urban-Rural Development of the People''s Republic of China, Circular of the General Office of the Ministry of Housing and Urban-Rural Development on the production safety accidents of housing and municipal engineering in 2019. https://www.mohurd.gov.cn/gongkai/fdzdgknr/tzgg/202006/20200624_246031.html, 2020 (accessed Feburary 14, 2022).

[7] W. Fang, L. Ding, P.E.D. Love, H. Luo, H. Li, F. Peña-Mora, B. Zhong, C. Zhou, Computer vision applications in construction safety assurance, Autom. Constr. 110 (2020), 103013, https://doi.org/10.1016/j.autcon.2019.103013.

[8] H. Luo, J. Liu, W. Fang, P.E. Love, Q. Yu, Z. Lu, Real-time smart video surveillance to manage safety: a case study of a transport mega-project, Adv. Eng. Inform. 45 (2020) 101100, https://doi.org/10.1016/j.aei.2020.101100.

[9] C. Zhou, H. Luo, W. Fang, R. Wei, L. Ding, Cyber-physical-system-based safety monitoring for blind hoisting with the internet of things: a case study, Autom. Constr. 97 (2019) 138–150, https://doi.org/10.1016/j.autcon.2018.10.017.

[10] S. Tang, D.R. Shelden, C.M. Eastman, P. Pishdad-Bozorgi, X. Gao, A review of building information modeling (BIM) and the internet of things (IoT) devices integration: present status and future trends, Autom. Constr. 101 (2019) 127–139, https://doi.org/10.1016/j.autcon.2019.01.020.

[11] C.J. Anumba, A. Akanmu, X. Yuan, C. Kan, Cyber—physical systems development for construction applications, Front. Eng. Manag. 8 (1) (2021) 72–87, https://doi.org/10.1007/s42524-020-0130-4.

[12] T. Hartmann, A. Trappey, Advanced engineering informatics-philosophical and methodological foundations with examples from civil and construction engineering, Develop. Built Environ. 4 (2020), 100020, https://doi.org/10.1016/j.dibe.2020.100020.

[13] Y. LeCun, Y. Bengio, G. Hinton, Deep learning, Nature 521 (7553) (2015) 436–444, https://doi.org/10.1038/nature14539.

[14] L. Ding, W. Fang, H. Luo, P.E. Love, B. Zhong, X. Ouyang, A deep hybrid learning model to detect unsafe behavior: integrating convolution neural networks and long short-term memory, Autom. Constr. 86 (2018) 118–124, https://doi.org/10.1016/j.autcon.2017.11.002.

[15] Q. Fang, H. Li, X. Luo, L. Ding, H. Luo, T.M. Rose, W. An, Detecting non-hardhat-use by a deep learning method from far-field surveillance videos, Autom. Constr. 85 (2018) 1–9, https://doi.org/10.1016/j.autcon.2017.09.018.

[16] B. Zhong, H. Wu, L. Ding, P.E. Love, H. Li, H. Luo, L. Jiao, Mapping computer vision research in construction: developments, knowledge gaps and implications for research, Autom. Constr. 107 (2019) 102919, https://doi.org/10.1016/j.autcon.2019.102919.

[17] T.D. Akinosho, L.O. Oyedele, M. Bilal, A.O. Ajayi, M.D. Delgado, O.O. Akinade, A. A. Ahmed, Deep learning in the construction industry: a review of present status and future innovations, J. Build. Eng. 32 (2020), 101827, https://doi.org/10.1016/j.jobe.2020.101827.

[18] L. Hou, H. Chen, G.K. Zhang, X. Wang, Deep learning-based applications for safety management in the AEC industry: a review, Appl. Sci. 11 (2) (2021) 821, https://doi.org/10.3390/app11020821.

[19] A. Pal, S.-H. Hsieh, Deep-learning-based visual data analytics for smart construction management, Autom. Constr. 131 (2021), 103892, https://doi.org/10.1016/j.autcon.2021.103892.

[20] A. Esteva, A. Robicquet, B. Ramsundar, V. Kuleshov, M. DePristo, K. Chou, C. Cui, G. Corrado, S. Thrun, J. Dean, A guide to deep learning in healthcare, Nat. Med. 25 (1) (2019) 24–29, https://doi.org/10.1038/s41591-018-0316-z.

[21] S. Ramos, S. Gehrig, P. Pinggera, U. Franke, C. Rother, Detecting unexpected obstacles for self-driving cars: Fusing deep learning and geometric modeling, in: 2017 IEEE Intelligent Vehicles Symposium (IV), 2017, pp. 1025–1032, https://doi.org/10.1109/IVS.2017.7995849.

[22] D.W. Otter, J.R. Medina, J.K. Kalita, A survey of the usages of deep learning for natural language processing, IEEE Trans. Neural Networks Learn. Syst. 32 (2) (2020) 604–624, https://doi.org/10.1109/TNNLS.2020.2979670.

[23] K. Gopalakrishnan, S.K. Khaitan, A. Choudhary, A. Agrawal, Deep convolutional neural networks with transfer learning for computer vision-based data-driven pavement distress detection, Constr. Build. Mater. 157 (2017) 322–330, https://doi.org/10.1016/j.conbuildmat.2017.09.110.

[24] L. Torrey, J. Shavlik, Transfer learning, handbook of research on machine learning applications and trends: algorithms, methods, and techniques, IGI Global (2010) 242–264, https://doi.org/10.4018/978-1-60566-766-9.ch011.

[25] S.J. Pan, Q. Yang, A survey on transfer learning, IEEE Trans. Knowl. Data Eng. 22 (10) (2009) 1345–1359, https://doi.org/10.1109/TKDE.2009.191.

[26] J. Han, S. Kang, Active learning with missing values considering imputation uncertainty, Knowl.-Based Syst. 224 (2021), 107079, https://doi.org/10.1016/j.knosys.2021.107079.

[27] J. Kim, J. Hwang, S. Chi, J. Seo, Towards database-free vision-based monitoring on construction sites: a deep active learning approach, Autom. Constr. 120 (2020), 103376, https://doi.org/10.1016/j.autcon.2020.103376.

[28] C. Shorten, T.M. Khoshgoftaar, A survey on image data augmentation for deep learning, J. Big Data 6 (1) (2019) 1–48, https://doi.org/10.1186/s40537-019-0197-0.

[29] W. Fang, P.E. Love, H. Luo, L. Ding, Computer vision for behaviour-based safety in construction: a review and future directions, Adv. Eng. Inform. 43 (2020) 100980, https://doi.org/10.1016/j.aei.2019.100980.

[30] A. Esteva, B. Kuprel, R.A. Novoa, J. Ko, S.M. Swetter, H.M. Blau, S. Thrun, Dermatologist-level classification of skin cancer with deep neural networks, Nature 542 (7639) (2017) 115–118, https://doi.org/10.1038/nature21056.

[31] H.A. Haenssle, C. Fink, R. Schneiderbauer, F. Toberer, T. Buhl, A. Blum, A. Kalloo, A.B.H. Hassen, L. Thomas, A. Enk, Man against machine: diagnostic performance of a deep learning convolutional neural network for dermoscopic melanoma recognition in comparison to 58 dermatologists, Ann. Oncol. 29 (8) (2018) 1836–1842, https://doi.org/10.1093/annonc/mdy166.

[32] A. Jamaludin, T. Kadir, A. Zisserman, SpineNet: automatically pinpointing classification evidence in spinal MRIs, in: International Conference on Medical Image Computing and Computer-Assisted Intervention, Springer, 2016, pp. 166–175, https://doi.org/10.1007/978-3-319-46723-8_20.

[33] S. Ren, K. He, R. Girshick, J. Sun, Faster r-cnn: towards real-time object detection with region proposal networks, IEEE Trans. Pattern Anal. Machine Intell. 39 (6) (2017) 1137–1149, https://doi.org/10.1109/TPAMI.2016.2577031.

[34] J. Dai, Y. Li, K. He, J. Sun, R-fcn: object detection via region-based fully convolutional networks, Adv. Neural Inf. Proces. Syst. 29 (2016), in: https://proceedings.neurips.cc/paper/2016/file/577ef1154f3240ad5b9b413aa7346a1e-Paper.pdf.

[35] W. Liu, D. Anguelov, D. Erhan, C. Szegedy, S. Reed, C.-Y. Fu, A.C. Berg, Ssd: single shot multibox detector, in: European Conference on Computer Vision, Springer, 2016, pp. 21–37, https://doi.org/10.1007/978-3-319-46448-0_2.

[36] J. Redmon, S. Divvala, R. Girshick, A. Farhadi, You only look once: unified, real-time object detection, Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (2016) 779–788, doi:110.1109/CVPR.2016.91.

[37] T.-Y. Lin, P. Goyal, R. Girshick, K. He, P. Dollár, Focal loss for dense object detection, in: Proceedings of the IEEE International Conference on Computer Vision, 2017, pp. 2980–2988. https://openaccess.thecvf.com/content_ICCV_2017/papers/Lin_Focal_Loss_for_ICCV_2017_paper.pdf.

[38] M. Danelljan, A. Robinson, F.S. Khan, M. Felsberg, Beyond correlation filters: learning continuous convolution operators for visual tracking, in: European Conference on Computer Vision, Springer, 2016, pp. 472–488, https://doi.org/10.1007/978-3-319-46454-1_29.

[39] N. Wang, D.Y. Yeung, Learning a deep compact image representation for visual tracking, Adv. Neural Inf. Proces. Syst. 26 (2013) 809–817, in: https://proceedings.neurips.cc/paper/2013/file/dc6a6489640ca02b0d42dabeb8e46bb7-Paper.pdf.

[40] L. Wang, W. Ouyang, X. Wang, H. Lu, Visual tracking with fully convolutional networks, in: Proceedings of the IEEE International Conference on Computer Vision, 2015, pp. 3119–3127. https://openaccess.thecvf.com/content_iccv_2015/papers/Wang_Visual_Tracking_With_ICCV_2015_paper.pdf.

[41] H. Nam, B. Han, Learning multi-domain convolutional neural networks for visual tracking, Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (2016) 4293–4302. https://openaccess.thecvf.com/content_cvpr_2016/papers/Nam_Learning_Multi-Domain_Convolutional_CVPR_2016_paper.pdf.

[42] K. He, G. Gkioxari, P. Dollár, R. Girshick, Mask r-cnn, in: Proceedings of the IEEE International Conference on Computer Vision, 2017, pp. 2961–2969. https://openaccess.thecvf.com/content_ICCV_2017/papers/He_Mask_R-CNN_ICCV_2017_paper.pdf.

[43] L.-C. Chen, A. Hermans, G. Papandreou, F. Schroff, P. Wang, H. Adam, Masklab: instance segmentation by refining object detection with semantic and direction features, Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (2018) 4013–4022. https://openaccess.thecvf.com/content_cvpr_2018/papers/Chen_MaskLab_Instance_Segmentation_CVPR_2018_paper.pdf.

[44] D. Bolya, C. Zhou, F. Xiao, Y.J. Lee, Yolact: real-time instance segmentation, in: Proceedings of the IEEE/CVF International Conference on Computer Vision, 2019, pp. 9157–9166. http://openaccess.thecvf.com/content_ICCV_2019/papers/Bolya_YOLACT_Real-Time_Instance_Segmentation_ICCV_2019_paper.pdf.

[45] S.-E. Wei, V. Ramakrishna, T. Kanade, Y. Sheikh, Convolutional pose machines, Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (2016) 4724–4732. https://openaccess.thecvf.com/content_cvpr_2016/papers/Wei_Convolutional_Pose_Machines_CVPR_2016_paper.pdf.

[46] A. Newell, K. Yang, J. Deng, Stacked hourglass networks for human pose estimation, in: European Conference on Computer Vision, Springer, 2016, pp. 483–499, https://doi.org/10.1007/978-3-319-46484-8_29.

[47] Y. Chen, Z. Wang, Y. Peng, Z. Zhang, G. Yu, J. Sun, Cascaded pyramid network for multi-person pose estimation, Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (2018) 7103–7112. https://openaccess.thecvf.com/content_cvpr_2018/papers/Chen_Cascaded_Pyramid_Network_CVPR_2018_paper.pdf.

[48] K. Sun, B. Xiao, D. Liu, J. Wang, Deep high-resolution representation learning for human pose estimation, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2019, pp. 5693–5703. https://openaccess.thecvf.com/content_CVPR_2019/papers/Sun_Deep_High-Resolution_Representation_Learning_for_Human_Pose_Estimation_CVPR_2019_paper.pdf.

[49] C. Zhu, F. Chen, Z. Shen, M. Savvides, Soft anchor-point object detection, in: Computer Vision–ECCV 2020: 16th European conference, Glasgow, UK, August 23–28, 2020, Proceedings, Part IX 16, 2020, pp. 91–107, https://doi.org/10.1007/978-3-030-58545-7_6.

[50] Z. Tian, C. Shen, H. Chen, T. He, Fcos: fully convolutional one-stage object detection, in: Proceedings of the IEEE/CVF International Conference on Computer Vision, 2019, pp. 9627–9636. https://openaccess.thecvf.com/content_ICCV_2019/papers/Tian_FCOS_Fully_Convolutional_One-Stage_Object_Detection_ICCV_2019_paper.pdf.

[51] C. Zhu, Y. He, M. Savvides, Feature selective anchor-free module for single-shot object detection, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2019, pp. 840–849. https://openaccess.thecvf.com/content_CVPR_2019/papers/Zhu_Feature_Selective_Anchor-Free_Module_for_Single-Shot_Object_Detection_CVPR_2019_paper.pdf.

[52] J. He, Z. Huang, N. Wang, Z. Zhang, Learnable graph matching: incorporating graph partitioning with deep feature learning for multiple object tracking, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2021, pp. 5299–5309. https://openaccess.thecvf.com/content/CVPR2021/papers/He_Learnable_Graph_Matching_Incorporating_Graph_Partitioning_With_Deep_Feature_Learning_CVPR_2021_paper.pdf.

[53] B. Shuai, A. Berneshawi, X. Li, D. Modolo, J. Tighe, SiamMOT: Siamese multi-object tracking, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2021, pp. 12372–12382. https://openaccess.thecvf.com/content/CVPR2021/papers/Shuai_SiamMOT_Siamese_Multi-Object_Tracking_CVPR_2021_paper.pdf.

[54] K.G. Quach, P. Nguyen, H. Le, T.D. Truong, C.N. Duong, M.T. Tran, K. Luu, DyGLIP: a dynamic graph model with link prediction for accurate multi-camera multiple object tracking, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2021, pp. 13784–13793. https://openaccess.thecvf.com/content/CVPR2021/papers/Quach_DyGLIP_A_Dynamic_Graph_Model_With_Link_Prediction_for_Accurate_CVPR_2021_paper.pdf.

[55] X. Zhang, H. Xu, H. Mo, J. Tan, C. Yang, L. Wang, W. Ren, Dcnas: densely connected neural architecture search for semantic image segmentation, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2021, pp. 13956–13967. https://openaccess.thecvf.com/content/CVPR2021/papers/Zhang_DCNAS_Densely_Connected_Neural_Architecture_Search_for_Semantic_Image_Segmentation_CVPR_2021_paper.pdf.

[56] Y. Yang, B. Lai, S. Soatto, DyStaB: Unsupervised object segmentation via dynamic-static bootstrapping, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2021, pp. 2826–2836. https://openaccess.thecvf.com/content/CVPR2021/papers/Yang_DyStaB_Unsupervised_Object_Segmentation_via_Dynamic-Static_Bootstrapping_CVPR_2021_paper.pdf.

[57] Y. Yuan, S.E. Wei, T. Simon, K. Kitani, J. Saragih, SimPoE: simulated character control for 3D human pose estimation, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2021, pp. 7159–7169. https://openaccess.thecvf.com/content/CVPR2021/papers/Yuan_SimPoE_Simulated_Character_Control_for_3D_Human_Pose_Estimation_CVPR_2021_paper.pdf.

[58] W. Mao, Z. Tian, X. Wang, C. Shen, FCPose: fully convolutional multi-person pose estimation with dynamic instance-aware convolutions, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2021, pp. 9034–9043. https://openaccess.thecvf.com/content/CVPR2021/papers/Mao_FCPose_Fully_Convolutional_Multi-Person_Pose_Estimation_With_Dynamic_Instance-Aware_Convolutions_CVPR_2021_paper.pdf.

[59] N. Indurkhya, F.J. Damerau, Handbook of Natural Language Processing, CRC Press, 2010, https://doi.org/10.1201/9781420085938.

[60] Y. Zou, A. Kiviniemi, S.W. Jones, Retrieving similar cases for construction project risk management using natural language processing techniques, Autom. Constr. 80 (2017) 66–76, https://doi.org/10.1016/j.autcon.2017.04.003.

[61] J. Devlin, M.-W. Chang, K. Lee, K. Toutanova, Bert: Pre-training of Deep Bidirectional Transformers for Language Understanding, arXiv preprint. arXiv:1810.04805, 2018. https://arxiv.org/pdf/1810.04805.pdf.

[62] Z. Huang, W. Xu, K. Yu, Bidirectional LSTM-CRF Models for Sequence Tagging, arXiv Preprint. arXiv:1508.01991, 2015. https://arxiv.org/pdf/1508.01991.pdf.

[63] J. Guo, Y. Fan, Q. Ai, W.B. Croft, A deep relevance matching model for ad-hoc retrieval, in: Proceedings of the 25th ACM International on Conference on Information and Knowledge Management, 2016, pp. 55–64, https://doi.org/10.1145/2983323.2983769.

[64] H. Zamani, M. Dehghani, W.B. Croft, E. Learned-Miller, J. Kamps, From neural re-ranking to neural ranking: Learning a sparse representation for inverted indexing, in: Proceedings of the 27th ACM International Conference on Information and Knowledge Management, 2018, pp. 497–506, https://doi.org/10.1145/3269206.3271800.

[65] W.M. Bramer, M.K. Rethlefsen, J. Kleijnen, O.H. Franco, Optimal database combinations for literature searches in systematic reviews: a prospective exploratory study, Syst. Rev. 6 (1) (2017) 1–12, https://doi.org/10.1186/s13643-017-0644-y.

[66] H.W. Heinrich, Industrial Accident Prevention. A Scientific Approach, Industrial Accident Prevention. A Scientific Approach, Second edition, 1941, https://doi.org/10.2105/ajph.22.1.119-b.

[67] H. Li, M. Lu, S.-C. Hsu, M. Gray, T. Huang, Proactive behavior-based safety management for construction safety improvement, Saf. Sci. 75 (2015) 107–117, https://doi.org/10.1016/j.ssci.2015.01.013.

[68] H. Li, X. Li, X. Luo, J. Siebert, Investigation of the causality patterns of non-helmet use behavior of construction workers, Autom. Constr. 80 (2017) 95–103, https://doi.org/10.1016/j.autcon.2017.02.006.

[69] N.D. Nath, A.H. Behzadan, S.G. Paal, Deep learning for site safety: real-time detection of personal protective equipment, Autom. Constr. 112 (2020), 103085, https://doi.org/10.1016/j.autcon.2020.103085.

[70] J. Shen, X. Xiong, Y. Li, W. He, P. Li, X. Zheng, Detecting safety helmet wearing on construction sites with bounding-box regression and deep transfer learning, Comp. Aided Civil Infrastruct. Eng. 36 (2) (2021) 180–196, https://doi.org/10.1111/mice.12579.

[71] J. Wu, N. Cai, W. Chen, H. Wang, G. Wang, Automatic detection of hardhats worn by construction personnel: a deep learning approach and benchmark dataset, Autom. Constr. 106 (2019), 102894, https://doi.org/10.1016/j.autcon.2019.102894.

[72] Q. Fang, H. Li, X. Luo, L. Ding, H. Luo, C. Li, Computer vision aided inspection on falling prevention measures for steeplejacks in an aerial environment, Autom. Constr. 93 (2018) 148–164, https://doi.org/10.1016/j.autcon.2018.05.022.

[73] W. Fang, B. Zhong, N. Zhao, P.E. Love, H. Luo, J. Xue, S. Xu, A deep learning-based approach for mitigating falls from height with computer vision: convolutional neural network, Adv. Eng. Inform. 39 (2019) 170–177, https://doi.org/10.1016/j.aei.2018.12.005.

[74] S. Bang, Y. Hong, H. Kim, Proactive proximity monitoring with instance segmentation and unmanned aerial vehicle-acquired video-frame prediction, Comp. Aided Civil Infrastruct. Eng. 36 (6) (2021) 800–816, https://doi.org/10.1111/mice.12672.

[75] D. Kim, M. Liu, S. Lee, V.R. Kamat, Remote proximity monitoring between mobile construction resources using camera-mounted UAVs, Autom. Constr. 99 (2019) 168–182, https://doi.org/10.1016/j.autcon.2018.12.014.

[76] X. Yan, H. Zhang, H. Li, Computer vision-based recognition of 3D relationship between construction entities for monitoring struck-by accidents, Comp. Aided Civil Infrastruct. Eng. 35 (9) (2020) 1023–1038, https://doi.org/10.1111/mice.12536.

[77] S. Han, S. Lee, F. Peña-Mora, Comparative study of motion features for similarity-based modeling and classification of unsafe actions in construction, J. Comput. Civ. Eng. 28 (5) (2014) A4014005, https://doi.org/10.1061/(ASCE)CP.1943-5487.0000339.

[78] X. Luo, H. Li, X. Yang, Y. Yu, D. Cao, Capturing and understanding workers' activities in far-field surveillance videos with deep action recognition and Bayesian nonparametric learning, Comp. Aided Civil Infrastruct. Eng. 34 (4) (2019) 333–351, https://doi.org/10.1111/mice.12419.

[79] D. Roberts, W.T. Calderon, S. Tang, M. Golparvar-Fard, Vision-based construction worker activity analysis informed by body posture, J. Comput. Civ. Eng. 34 (4) (2020) 04020017, https://doi.org/10.1061/(ASCE)CP.1943-5487.0000898.

[80] W. Fang, L. Ma, P.E. Love, H. Luo, L. Ding, A. Zhou, Knowledge graph for identifying hazards on construction sites: integrating computer vision with ontology, Autom. Constr. 119 (2020) 103310, https://doi.org/10.1016/j.autcon.2020.103310.

[81] Z. Pan, C. Su, Y. Deng, J. Cheng, Video2Entities: a computer vision-based entity extraction framework for updating the architecture, engineering and construction industry knowledge graphs, Autom. Constr. 125 (2021), 103617, https://doi.org/10.1016/j.autcon.2021.103617.

[82] R. Xiong, Y. Song, H. Li, Y. Wang, Onsite video mining for construction hazards identification with visual relationships, Adv. Eng. Inform. 42 (2019), 100966, https://doi.org/10.1016/j.aei.2019.100966.

[83] B. Zhong, H. Li, H. Luo, J. Zhou, W. Fang, X. Xing, Ontology-based semantic modeling of knowledge in construction: classification and identification of hazards implied in images, J. Constr. Eng. Manag. 146 (4) (2020) 04020013, https://doi.org/10.1061/(ASCE)CO.1943-7862.0001767.

[84] S. Tang, D. Roberts, M. Golparvar-Fard, Human-object interaction recognition for automatic construction site safety inspection, Autom. Constr. 120 (2020), 103356, https://doi.org/10.1016/j.autcon.2020.103356.

[85] W. Fang, L. Ding, H. Luo, P.E. Love, Falls from heights: a computer vision-based approach for safety harness detection, Autom. Constr. 91 (2018) 53–61, https://doi.org/10.1016/j.autcon.2018.02.018.

[86] H. Son, H. Seong, H. Choi, C. Kim, Real-time vision-based warning system for prevention of collisions between workers and heavy equipment, J. Comput. Civ. Eng. 33 (5) (2019) 04019029, https://doi.org/10.1061/(ASCE)CP.1943-5487.0000845.

[87] Z.H. Lin, A.Y. Chen, S.H. Hsieh, Temporal image analytics for abnormal construction activity identification, Autom. Constr. 124 (2021), 103572, https://doi.org/10.1016/j.autcon.2021.103572.

[88] H. Wu, B. Zhong, H. Li, P. Love, X. Pan, N. Zhao, Combining computer vision with semantic reasoning for on-site safety management in construction, J. Build. Eng. 42 (2021), 103036, https://doi.org/10.1016/j.jobe.2021.103036.

[89] D. Balageas, C.-P. Fritzen, A. Güemes, Structural Health Monitoring, John Wiley & Sons, 2010. ISBN:978-1-905209-01-9.

[90] S. Park, S. Bang, H. Kim, H. Kim, Patch-based crack detection in black box images using convolutional neural networks, J. Comput. Civ. Eng. 33 (3) (2019) 04019017, https://doi.org/10.1061/(ASCE)CP.1943-5487.0000831.

[91] D. Kang, Y.-J. Cha, Autonomous UAVs for structural health monitoring using deep learning and an ultrasonic beacon system with geo-tagging, Comp. Aided Civil Infrastruct. Eng. 33 (10) (2018) 885–902, https://doi.org/10.1111/mice.12375.

[92] C. Zhang, C.-C. Chang, M. Jamshidi, Concrete bridge surface damage detection using a single-stage detector, Comp. Aided Civil Infrastruct. Eng. 35 (4) (2020) 389–409, https://doi.org/10.1111/mice.12500.

[93] S. Bang, S. Park, H. Kim, H. Kim, Encoder–decoder network for pixel-level road crack detection in black-box images, Comp. Aided Civil Infrastruct. Eng. 34 (8) (2019) 713–727, https://doi.org/10.1111/mice.12440.

[94] K. Jang, Y.-K. An, B. Kim, S. Cho, Automated crack evaluation of a high-rise bridge pier using a ring-type climbing robot, Comp. Aided Civil Infrastruct. Eng. 36 (1) (2021) 14–29, https://doi.org/10.1111/mice.12550.

[95] S. Jiang, J. Zhang, Real-time crack assessment using deep neural networks with wall-climbing unmanned aerial system, Comp. Aided Civil Infrastruct. Eng. 35 (6) (2020) 549–564, https://doi.org/10.1111/mice.12519.

[96] R. Kalfarisi, Z.Y. Wu, K. Soh, Crack detection and segmentation using deep learning with 3D reality mesh model for quantitative assessment and integrated visualization, J. Comput. Civ. Eng. 34 (3) (2020) 04020010, https://doi.org/10.1061/(ASCE)CP.1943-5487.0000890.

[97] J. Liu, X. Yang, S. Lau, X. Wang, S. Luo, V.C.S. Lee, L. Ding, Automated pavement crack detection and segmentation based on two-step convolutional neural network, Comp. Aided Civil Infrastruct. Eng. 35 (11) (2020) 1291–1305, https://doi.org/10.1111/mice.12622.

[98] Q. Song, Y. Wu, X. Xin, L. Yang, M. Yang, H. Chen, C. Liu, M. Hu, X. Chai, J. Li, Real-time tunnel crack analysis system via deep learning, IEEE Access 7 (2019) 64186–64197, https://doi.org/10.1109/ACCESS.2019.2916330.

[99] X. Zhang, D. Rajan, B. Story, Concrete crack detection using context-aware deep semantic segmentation network, Comp. Aided Civil Infrastruct. Eng. 34 (11) (2019) 951–971, https://doi.org/10.1111/mice.12477.

[100] F. Wei, G. Yao, Y. Yang, Y. Sun, Instance-level recognition and quantification for concrete surface bughole based on deep learning, Autom. Constr. 107 (2019), 102920, https://doi.org/10.1016/j.autcon.2019.102920.

[101] H. Wu, L. Yao, Z. Xu, Y. Li, X. Ao, Q. Chen, Z. Li, B. Meng, Road pothole extraction and safety evaluation by integration of point cloud and images derived from mobile mapping sensors, Adv. Eng. Inform. 42 (2019), 100936, https://doi.org/10.1016/j.aei.2019.100936.

[102] D.J. Atha, M.R. Jahanshahi, Evaluation of deep learning approaches based on convolutional neural networks for corrosion detection, Struct. Health Monit. 17 (5) (2018) 1110–1128, doi:10.1177%2F1475921717737051.

[103] T.-C. Huynh, J.-H. Park, H.-J. Jung, J.-T. Kim, Quasi-autonomous bolt-loosening detection method using vision-based deep learning and image processing, Autom. Constr. 105 (2019), 102844, https://doi.org/10.1016/j.autcon.2019.102844.

[104] S.S. Kumar, D.M. Abraham, M.R. Jahanshahi, T. Iseley, J. Starr, Automated defect classification in sewer closed circuit television inspections using deep convolutional neural networks, Autom. Constr. 91 (2018) 273–283, https://doi.org/10.1016/j.autcon.2018.03.028.

[105] M. Wang, S.S. Kumar, J.C.P. Cheng, Automated sewer pipe defect tracking in CCTV videos based on defect detection and metric learning, Autom. Constr. 121 (2021), 103438, https://doi.org/10.1016/j.autcon.2020.103438.

[106] G. Pan, Y. Zheng, S. Guo, Y. Lv, Automatic sewer pipe defect semantic segmentation based on improved U-net, Autom. Constr. 119 (2020) 103383, https://doi.org/10.1016/j.autcon.2020.103383.

[107] Y. Jang, Y. Ahn, Y. Kim Ha, Estimating compressive strength of concrete using deep convolutional neural networks with digital microscope images, J. Comput. Civ. Eng. 33 (3) (2019) 04019018, https://doi.org/10.1061/(ASCE)CP.1943-5487.0000837.

[108] W. Wang, P. Shi, L. Deng, H. Chu, X. Kong, Residual strength evaluation of corroded textile-reinforced concrete by the deep learning-based method, Materials 13 (14) (2020) 3226, https://doi.org/10.3390/ma13143226.

[109] M. Azimi, G. Pekcan, Structural health monitoring using extremely compressed data through deep learning, Comp. Aided Civil Infrastruct. Eng. 35 (6) (2020) 597–614, https://doi.org/10.1111/mice.12517.

[110] Y.-Z. Lin, Z.-H. Nie, H.-W. Ma, Structural damage detection with automatic feature-extraction through deep learning, Comp. Aided Civil Infrastruct. Eng. 32 (12) (2017) 1025–1046, https://doi.org/10.1111/mice.12313.

[111] F. Ni, J. Zhang, M.N. Noori, Deep learning for data anomaly detection and data compression of a long-span suspension bridge, Comp. Aided Civil Infrastruct. Eng. 35 (7) (2020) 685–700, https://doi.org/10.1111/mice.12528.

[112] S.O. Sajedi, X. Liang, Vibration-based semantic damage segmentation for large-scale structural health monitoring, Comp. Aided Civil Infrastruct. Eng. 35 (6) (2020) 579–596, https://doi.org/10.1111/mice.12523.

[113] F. Zhang, A hybrid structured deep neural network with Word2Vec for construction accident causes classification, Int. J. Constr. Manag. (2019) 1–21, https://doi.org/10.1080/15623599.2019.1683692.

[114] J. Zhang, L. Zi, Y. Hou, D. Deng, W. Jiang, M. Wang, A C-BiLSTM approach to classify construction accident reports, Appl. Sci. (Switzerland) 10 (17) (2020) 5754, https://doi.org/10.3390/app10175754.

[115] H. Baker, M.R. Hallowell, A.J.P. Tixier, Automatically learning construction injury precursors from text, Autom. Constr. 118 (2020), 103145, https://doi.org/10.1016/j.autcon.2020.103145.

[116] W. Fang, H. Luo, S. Xu, P.E.D. Love, Z. Lu, C. Ye, Automated text classification of near-misses from safety reports: An improved deep learning approach, Adv. Eng. Inform. 44 (2020), 101060, https://doi.org/10.1016/j.aei.2020.101060.

[117] B. Zhong, X. Pan, P.E.D. Love, L. Ding, W. Fang, Deep learning and network analysis: classifying and visualizing accident narratives in construction, Autom. Constr. 113 (2020), 103089, https://doi.org/10.1016/j.autcon.2020.103089.

[118] B. Zhong, X. Pan, P.E.D. Love, J. Sun, C. Tao, Hazard analysis: a deep learning and text mining framework for accident prevention, Adv. Eng. Inform. 46 (2020), 101152, https://doi.org/10.1016/j.aei.2020.101152.
[119] D. Feng, H. Chen, A small samples training framework for deep learning-based automatic information extraction: case study of construction accident news reports analysis, Adv. Eng. Inform. 47 (2021), 101256, https://doi.org/10.1016/j.aei.2021.101256.
[120] D. Guo, J. Li, S.-H. Jiang, X. Li, Z. Chen, Intelligent assistant driving method for tunnel boring machine based on big data, Acta Geotech. (2021) 1–12, https://doi.org/10.1007/s11440-021-01327-1.
[121] Q. Wang, X. Xie, H. Yu, M.A. Mooney, Predicting slurry pressure balance with a long short-term memory recurrent neural network in difficult ground condition, Comp. Intel. Neurosci. 2021 (2021), https://doi.org/10.1155/2021/6678355.
[122] Y. Zhang, G. Gong, H. Yang, W. Li, J. Liu, Precision versus intelligence: autonomous supporting pressure balance control for slurry shield tunnel boring machines, Autom. Constr. 114 (2020), 103173, https://doi.org/10.1016/j.autcon.2020.103173.
[123] C. Zhou, H. Xu, L. Ding, L. Wei, Y. Zhou, Dynamic prediction for attitude and position in shield tunneling: a deep learning method, Autom. Constr. 105 (2019), 102840, https://doi.org/10.1016/j.autcon.2019.102840.
[124] W. Fang, P.E. Love, L. Ding, S. Xu, T. Kong, H. Li, Computer vision and deep learning to manage safety in construction: matching images of unsafe behavior and semantic rules, IEEE Trans. Eng. Manag. (2021), https://doi.org/10.1109/TEM.2021.3093166.
[125] D. Roberts, M. Golparvar-Fard, End-to-end vision-based detection, tracking and activity analysis of earthmoving equipment filmed at ground level, Autom. Constr. 105 (2019), 102811, https://doi.org/10.1016/j.autcon.2019.04.006.
[126] B. Xiao, S.-C. Kang, Development of an image data set of construction machines for deep learning object detection, J. Comput. Civ. Eng. 35 (2) (2021) 05020005, https://doi.org/10.1061/(ASCE)CP.1943-5487.0000945.
[127] C.R. Ahn, S. Lee, C. Sun, H. Jebelli, K. Yang, B. Choi, Wearable sensing technology applications in construction safety and health, J. Constr. Eng. Manag. 145 (11) (2019) 03119007, https://doi.org/10.1061/(ASCE)CO.1943-7862.0001708.
[128] I. Awolusi, E. Marks, M. Hallowell, Wearable technology for personalized construction safety monitoring and trending: review of applicable devices, Autom. Constr. 85 (2018) 96–106, https://doi.org/10.1016/j.autcon.2017.10.010.
[129] M.T. Musavi, K.H. Chan, D.M. Hummels, K. Kalantri, On the generalization ability of neural network classifiers, IEEE Trans. Pattern Anal. Mach. Intell. 16 (6) (1994) 659–663, https://doi.org/10.1109/34.295911.
[130] C. Fan, C. Zhang, A. Yahja, A. Mostafavi, Disaster City digital twin: a vision for integrating artificial and human intelligence for disaster management, Int. J. Inf. Manag. 56 (2021), 102049.